

NCAN • STRATTON VA MEDICAL CENTER • SUMMER 2026

Can the brain tell us how big the reflex will be?

Machine learning on 2.7 million trials of twenty-year-old neural recordings.

Suchith Rao • Tarun Senthil

with Dr. Jonathan Carp and Theresa Vaughan



Roadmap

- 1 The question** what conditioning is, and what was already known
- 2 The data** one trial, six animals, and a broken decoder
- 3 How we built it** the model, the stack, the daily workflow
- 4 What we found** two results, one negative, and the controls
- 5 Where it goes** what we'd publish, and what it's useful for

What the animals are doing

1

The reflex

A small electrical stimulus to the leg nerve produces a reflex in the soleus muscle. Its size is what we measure.

2

The training

The rat is rewarded whenever that reflex is bigger than a threshold — or smaller, depending on the group.

3

The learning

Over weeks the reflex shifts in the trained direction. A 20% change counts as success, and the long-term change is in the spinal cord.

What this builds on: Boulay, Chen & Wolpaw, 2015

J Neurophysiol 113: 2232–2241, 2015.
First published January 28, 2015; doi:10.1152/jn.00677.2014.

Electrocorticographic activity over sensorimotor cortex and motor function in awake behaving rats

Chadwick B. Boulay, Xiang Yang Chen, and Jonathan R. Wolpaw

Laboratory of Neural Injury and Repair, Wadsworth Center, New York State Department of Health, Albany, New York; and State University of New York, Albany, New York

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Boulay CB, Chen XY, Wolpaw JR. Electrocorticographic activity over sensorimotor cortex and motor function in awake behaving rats. *J Neurophysiol* 113: 2232–2241, 2015. First published January 28, 2015; doi:10.1152/jn.00677.2014.—Sensorimotor cortex exerts both short-term and long-term control over the spinal reflex pathways that serve motor behaviors. Better understanding of this control could offer new possibilities for restoring function after central nervous system trauma or disease. We examined the impact of ongoing sensorimotor cortex (SMC) activity on the largely monosynaptic pathway of the H-reflex, the electrical analog of the spinal stretch reflex. In 41 awake adult rats, we measured soleus electromyographic (EMG) activity, the soleus H-reflex, and electrocorticographic activity over the contralat-

aspects of motor function: the spinal cord reflex pathways that are part of the substrate supporting all motor behaviors. As a result, spinal cord reflexes are serving as models for studying CNS function and plasticity and as therapeutic targets in spinal cord injury, stroke, and other disorders (Chen et al. 2006c; Thompson et al. 2013; Wolpaw 2010).

The H-reflex is a particularly attractive experimental model because it is mediated by a wholly spinal and largely monosynaptic spinal pathway (Henneman et al. 1965; Matthews 1972). The motoneuron excitation produced by the H-reflex pathway contributes to locomotion and to other important

Their question

How does ongoing sensorimotor cortex activity affect H-reflex size? 41 awake rats; ECoG, soleus EMG and H-reflex recorded together.

What they found

- Cortical activity fell into three bands: $\mu\beta$ (5–30 Hz), low γ (40–85 Hz), high γ (100–200 Hz).
- Background soleus EMG tracked cortical state: lower $\mu\beta$ and higher low- γ power went with more muscle activity.
- H-reflex size related to the same bands in the opposite direction, but only once background EMG was in the model.

J Neurophysiol 113:2232–2241, 2015

Already known: cortex influences reflex size. learns?

Our question: is it visible on a single trial, and does it change as the animal

Why bring machine learning to this

The usual approach

Decide in advance which features of the recording might matter — a peak here, a band of power there — then measure those.

Whatever you didn't think of, you don't measure.

What we did instead

Let a network learn its own compact description of each trace, with no labels and no assumptions, then test what that description can predict.

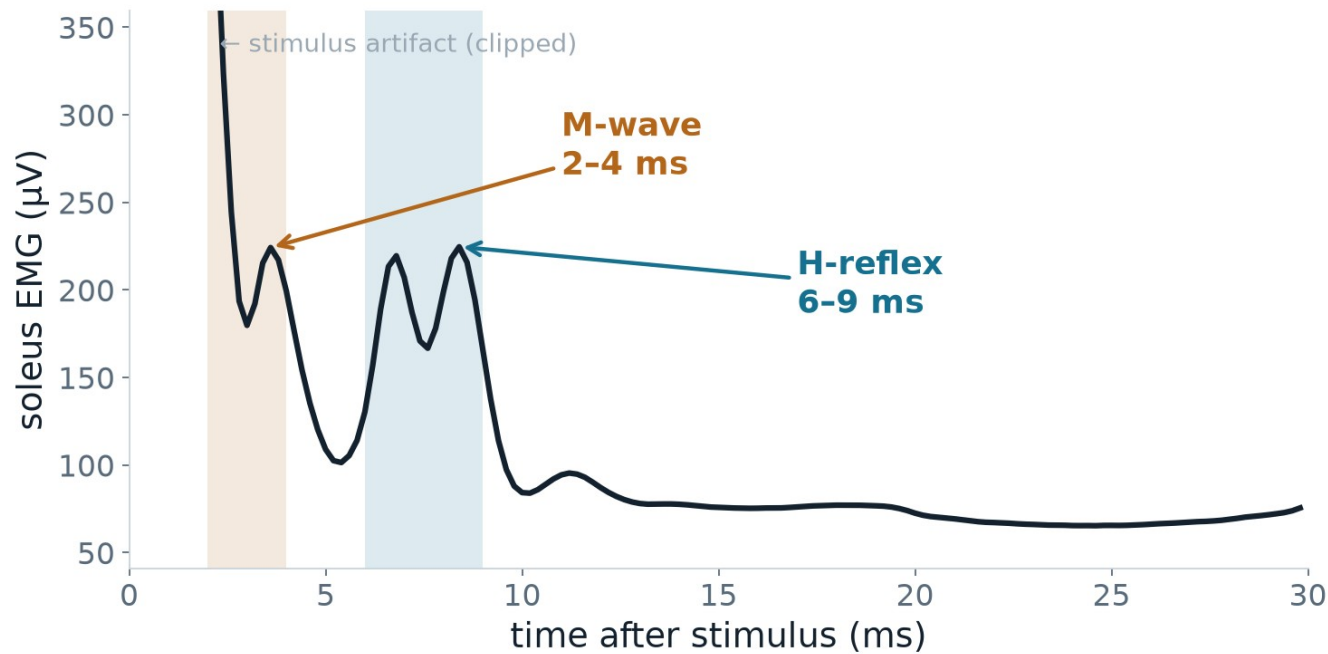
With 2.7 million trials, that is tractable.

The precedent: BrainBERT

BrainBERT (Wang et al., ICLR 2023) is a self-supervised model for human intracranial recordings: it learns a general representation of the neural signal first, without labels, and then uses it for downstream questions. We use a much smaller model, because we have one channel and a 30 ms window rather than a whole electrode array.

What a single trial looks like

Every stimulus produces one trial: a 30-millisecond window, recorded from a pair of EMG electrodes in the soleus muscle.



Reading the trace

- **The spike at zero**

the stimulus itself — an electrical artifact

- **M-wave, 2-4 ms**

the nerve firing the muscle directly

- **H-reflex, 6-9 ms**

the signal that travelled to the spinal cord and back — this is what conditioning changes

First problem: the archive would not decode

What we were given

2006 MySQL tables and a decoder script. The script said each sample's two bytes are stored most-significant first (big-endian).

What happened

Decoded as documented, every trial came out as noise. No M-wave, no reflex, nothing.

What was wrong

They are stored least-significant first (little-endian). Read that way, the physiology appears exactly where the experimenters' own notes said it would.

How we knew we had it right

Not because it looked plausible — because the M-wave and H-reflex landed at exactly the latencies the experimenter had typed into the log in 2006: “MR interval 2-4; HR interval 6-9”. The data confirmed the notes and the notes confirmed the data.

Six animals, 2.7 million trials

2.7M

trials decoded

6

animals

3 / 3

down / up

45–210

days each

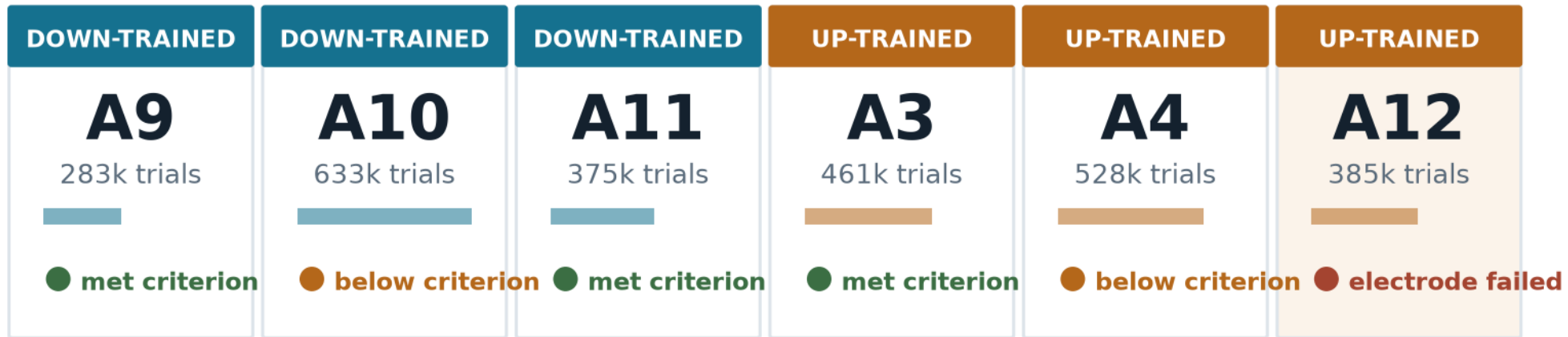
Choosing which animals to download, without downloading them

The log files are a few hundred kilobytes; the signal data is four gigabytes an animal. The experimenter raises the reward threshold whenever the rat exceeds it, so the threshold history records how the animal was doing. We read only the logs, then downloaded signal data for the animals that responded.

One animal excluded. A12's pre-stimulus cortical signal is 2.8 μV against 106–128 μV in the others — that electrode had failed.

Who is in the study

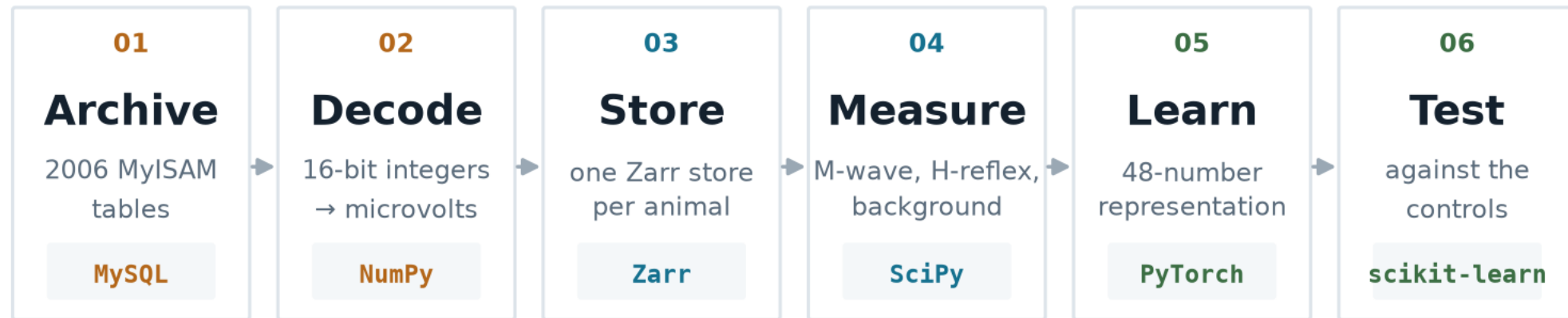
SIX ANIMALS PROCESSED



Three trained to increase the reflex, three to decrease it. Three met the 20% criterion; two did not; one electrode had failed.

The two below criterion act as a negative control later in the analysis.

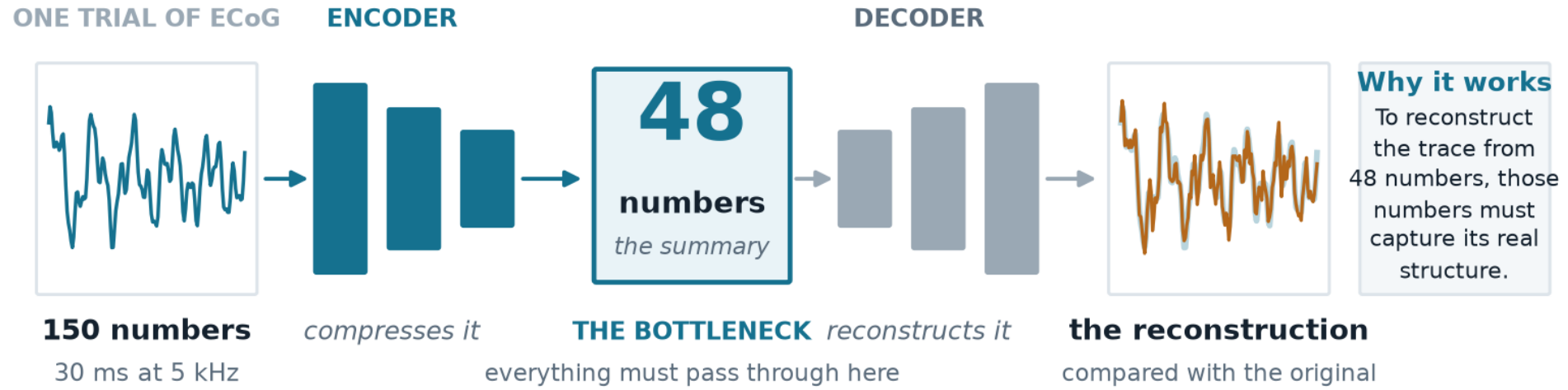
The pipeline, end to end



2.7 million trials · 6 animals · about 4 GB in per animal, 300 MB out · two commands per animal

Fully scripted and version-controlled. Re-running an animal from scratch takes minutes.

What the model actually is



The model is never told what to look for.

It never sees an H-reflex, a label, or a training phase. It only tries to reconstruct what it was given, so the 48 numbers reflect the structure of the cortical signal itself rather than the features we assumed would matter.

Training it, and the one trap we avoided

How it was trained

600,000	trials pooled across five animals
12	passes through the data
~4 min	on a laptop GPU
0.0149 → 0.0070	reconstruction error

The trap

Pool several animals naively and the model learns to tell the animals apart — different electrodes, different amplitudes. That is a recording artefact, not biology.

Fix: standardise every animal separately before pooling.

Why one model across all animals, rather than one per animal

A separate model per animal would describe each brain in its own private vocabulary, and you could never compare them. Training one model on all of them forces a shared vocabulary — so “this pattern of cortical activity” means the same thing in animal 3 as it does in animal 11. That is what makes the group comparisons later in the talk legitimate.

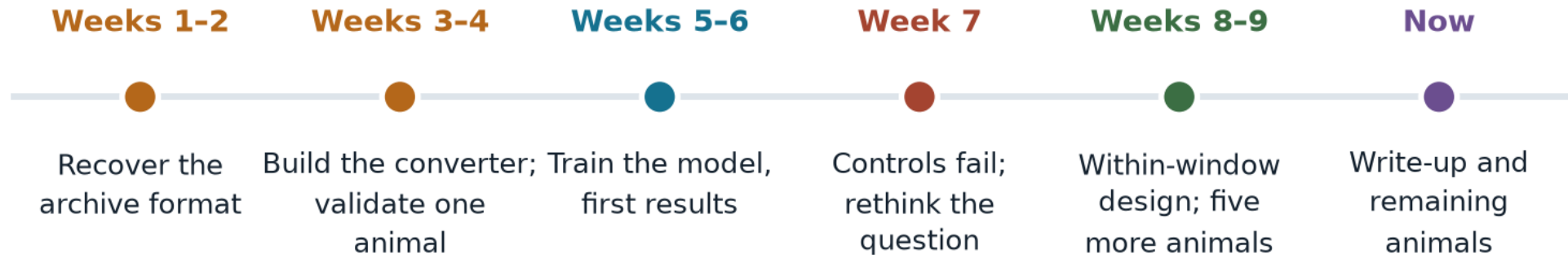
Tech stack and daily workflow

DATA	MODEL	ANALYSIS	DELIVERY
- MySQL / MariaDB - Zarr - NumPy	- PyTorch - Apple MPS / GPU 48-dim encoder	- scikit-learn - SciPy - ridge + permutation	- Git / GitHub - matplotlib - automated figures

Daily loop: read the log → load one animal → convert → inspect the averaged trace → run the analyses → commit figures.

Every figure in this talk regenerates from a single command. None were hand-edited.

How the summer actually went



The week the controls failed was the most useful week of the project.

It removed the result we thought we had and produced the design the current finding rests on.

Why reflex size alone tells you nothing

The problem

A bigger stimulus makes a bigger reflex — no learning required — and so does a more active muscle. Boulay et al. (2015) found exactly this: H-reflex size varied with M-wave size and background EMG as well as with cortical activity.

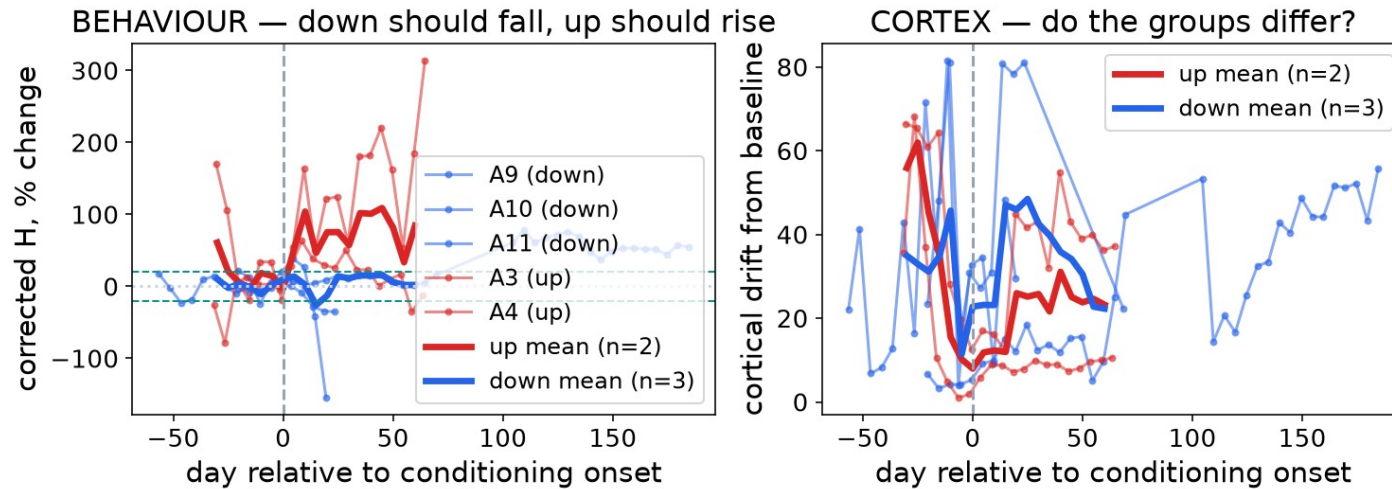
In one animal the delivered stimulus nearly quintupled over the recording; early on, that made one of our results come out backwards.

The fix

The M-wave sits in the same trace and reflects only the stimulus. It is a per-trial receipt for how much stimulus arrived. The 20 ms before each stimulus measures how active the muscle already was. We remove both from every measure.

Every number in the rest of this talk has both corrections applied.

Conditioning works, in both directions



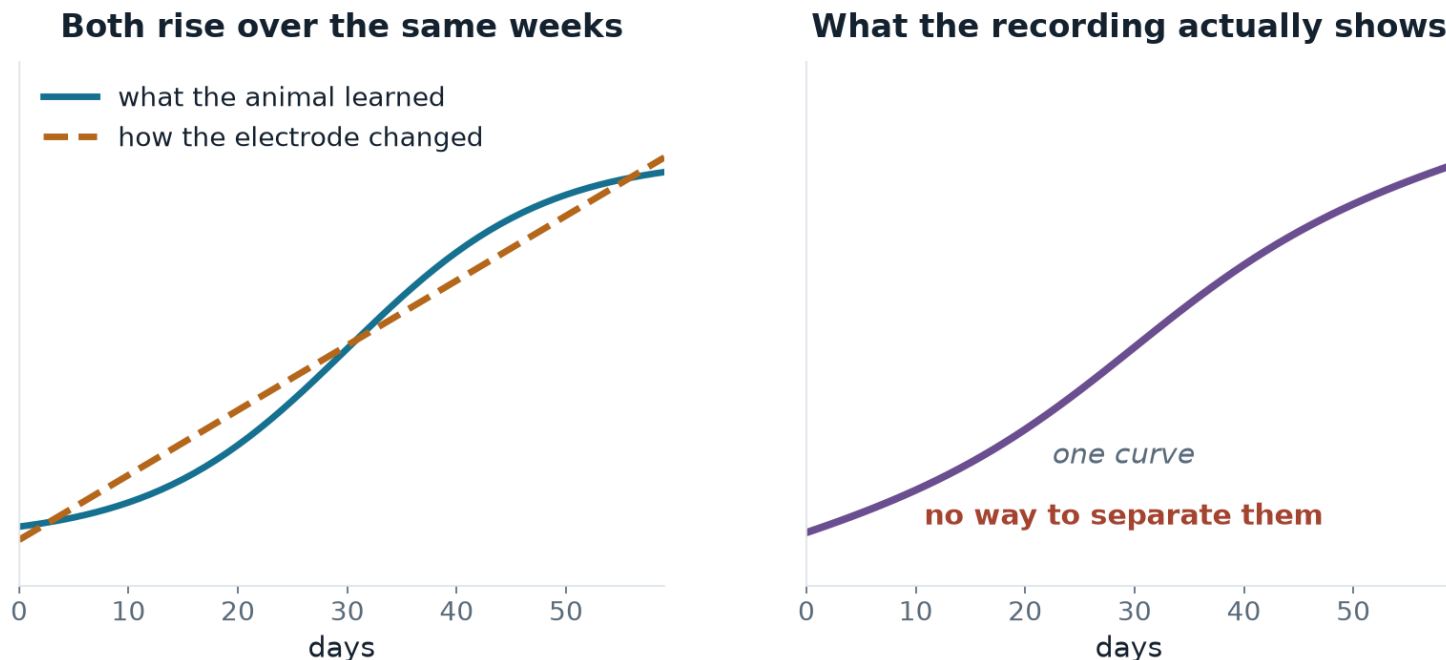
The positive control

Each animal is lined up to its own training start day. Up-trained animals (red) rise; down-trained (blue) fall.

Three of five clear the 20% criterion in the trained direction.

The two failures become a built-in control later on.

Two things change slowly over the same weeks



Within one animal, learning and electrode change are both smooth functions of time. That is why the first analysis could not work.

This is a limit of the study design, not of the data — and it shaped everything we did next.

Our first approach did not survive its controls

We measured how far the average brain state moved across training. It moves — but three tests say the movement is the recording, not the animal.

Take baseline-only data and pretend half was training

Nothing happened, so it should show no movement

Moves just as much — 5 of 6 animals

Shuffle which trial belongs to which day

Keeps the numbers, destroys the timeline

No movement — the measure itself is sound

Compare animals trained up against trained down

Electrode drift doesn't care which way you trained

No difference between the groups

These controls changed our conclusion.

So we asked a question that drift cannot fake

Instead of

“Did the average brain state move over weeks?”

We ask

“Can the brain predict this trial's reflex?”

THE WHOLE RECORDING



ONE FIVE-DAY WINDOW



WHY DRIFT CANNOT FAKE IT

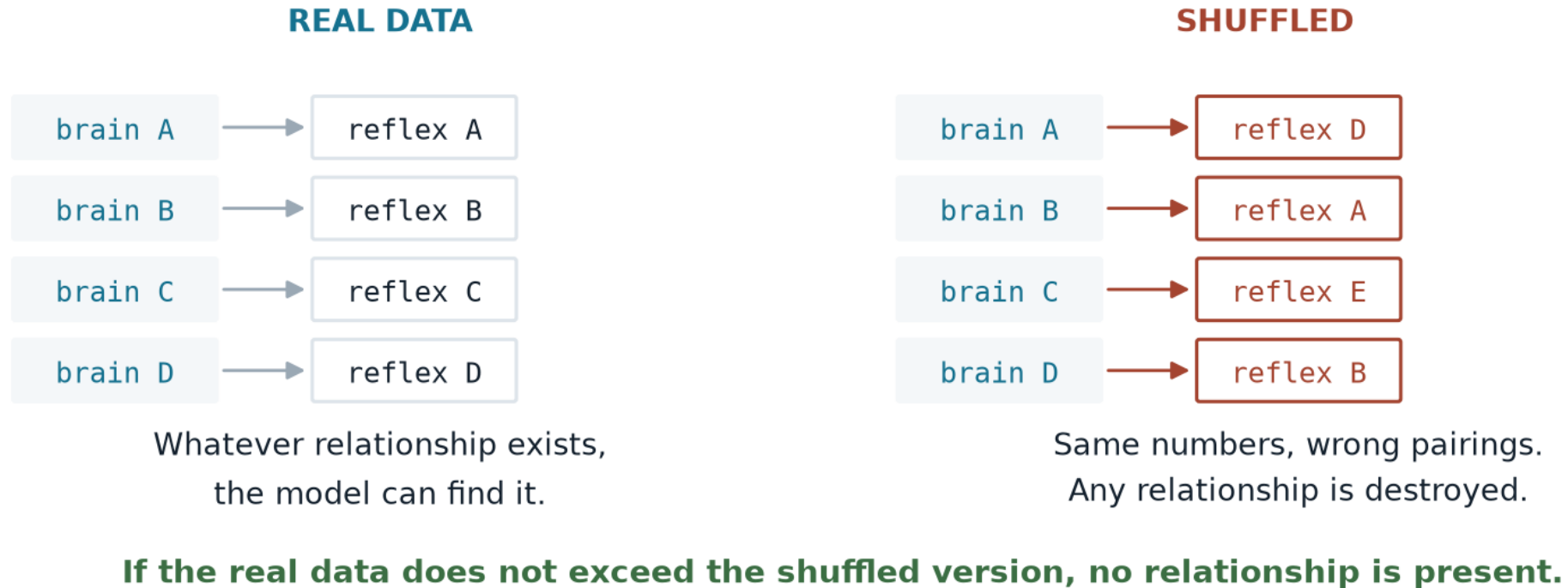
Within five days the electrode is essentially fixed.
Any remaining drift shifts training and test trials together, so it cancels.

Drift can move an average. It cannot create a trial-by-trial relationship inside five days.

Repeated in every window — 101 across five animals — each compared against a shuffle of its own trials.

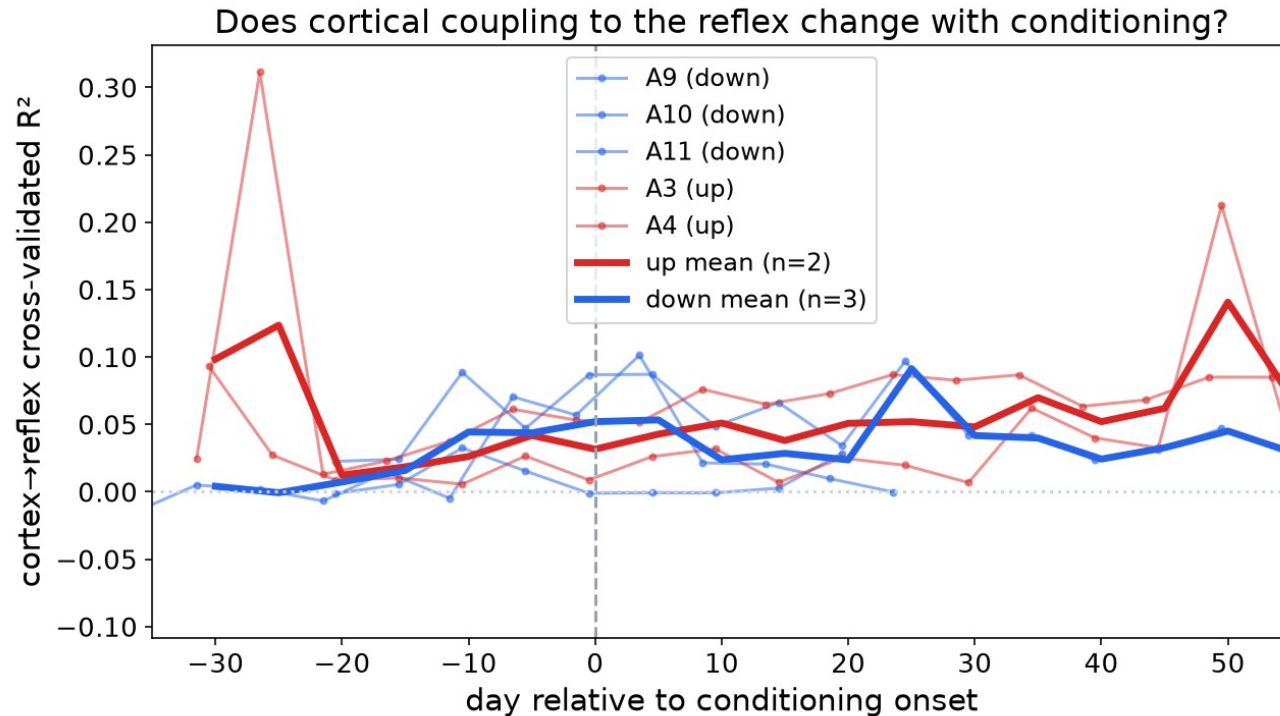
A window counts only if the real data exceeds its own shuffled counterpart.

And each window is judged against itself



We repeat this in all 101 windows across five animals. A window counts only if the real data beats its own shuffled version.

The brain does carry a signal about the reflex



99 of 101

windows where the real data beat its own shuffled version

Present in all five animals.

Counting the animal as the unit of analysis:

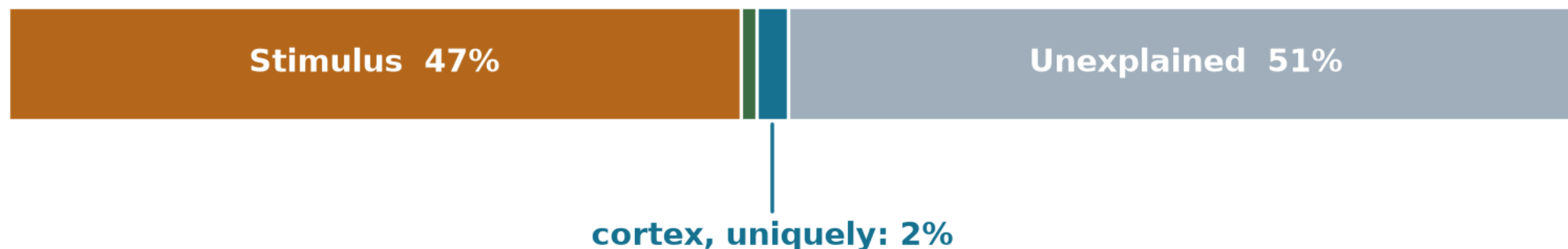
$t(4) = 3.81$, $p = 0.019$

Small in size, but almost never absent.

How much does the brain actually explain?

When parsing the variance in reflex size, the stimulus dominates, and the brain overlaps with it heavily.

Sources of variation in reflex size

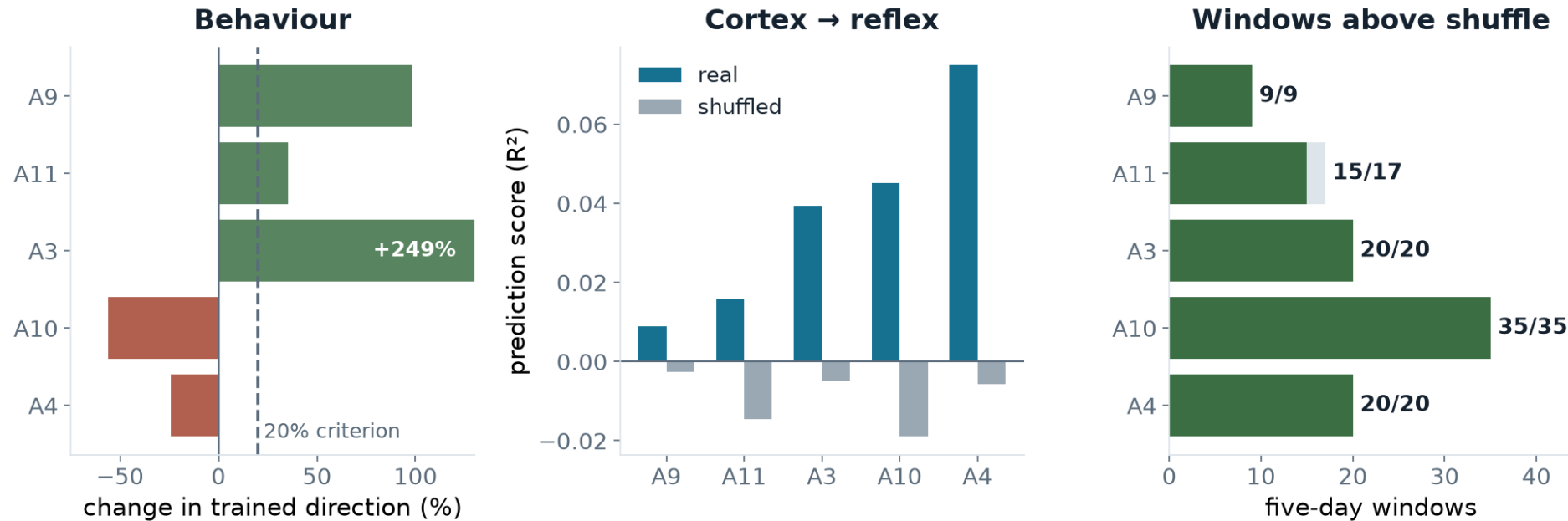


Cortex alone appears to explain 21%, but almost all of it overlaps with the stimulus, which drives both signals.

Why the brain's own 21% collapses to 2%

The stimulus drives both the cortical evoked response and the reflex, so most of what the brain appears to explain is really the stimulus. Only the part that survives once stimulus and background are already in the model counts as a genuine cortical contribution.

All five animals, all three measures



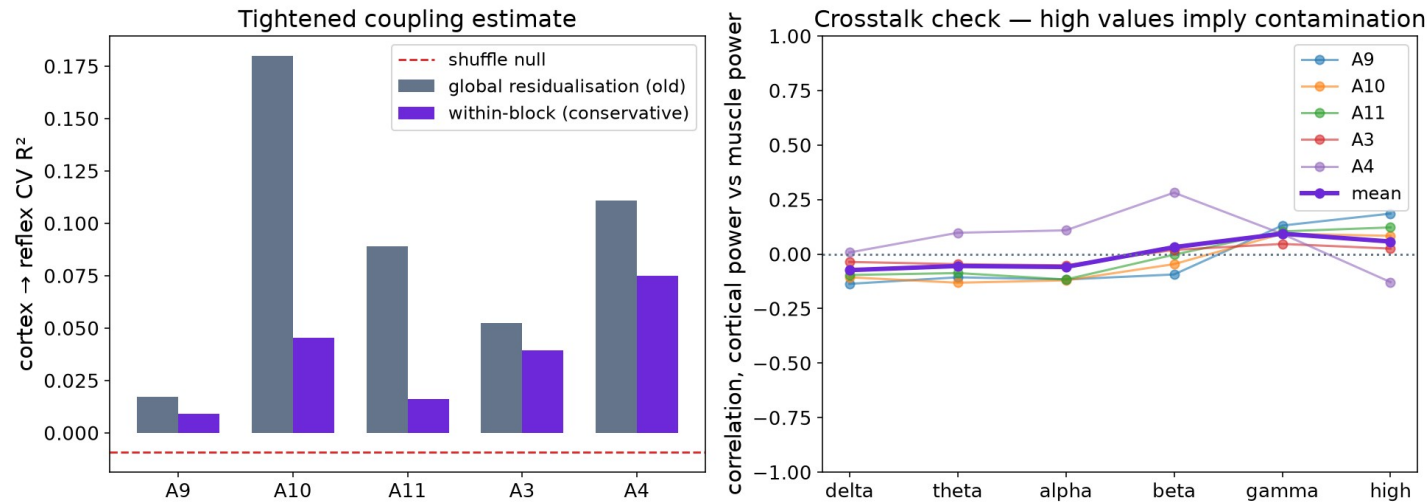
Left: three of five passed the 20% criterion; two moved against training. Middle: real prediction exceeds the shuffled control in every animal. Right: 99 of 101 windows.

Every control we applied

Baseline sham	Pretend part of the baseline was training	drift fails
Shuffled trial order	Keep the numbers, destroy the timeline	null ≈ 0
Stimulus strength	Is it just how strong the stimulus was?	survives
Pre-stimulus background	Is it muscle tone rather than brain?	survives
Within-window testing	Could electrode drift manufacture it?	drift cancels
Muscle crosstalk	Is the brain electrode picking up EMG?	$r = +0.06$
Up versus down	Does it depend on training direction?	no difference

Five it survives, two it fails — and the two failures are what told us the average-state approach was the wrong question.

Is the brain electrode just picking up muscle?



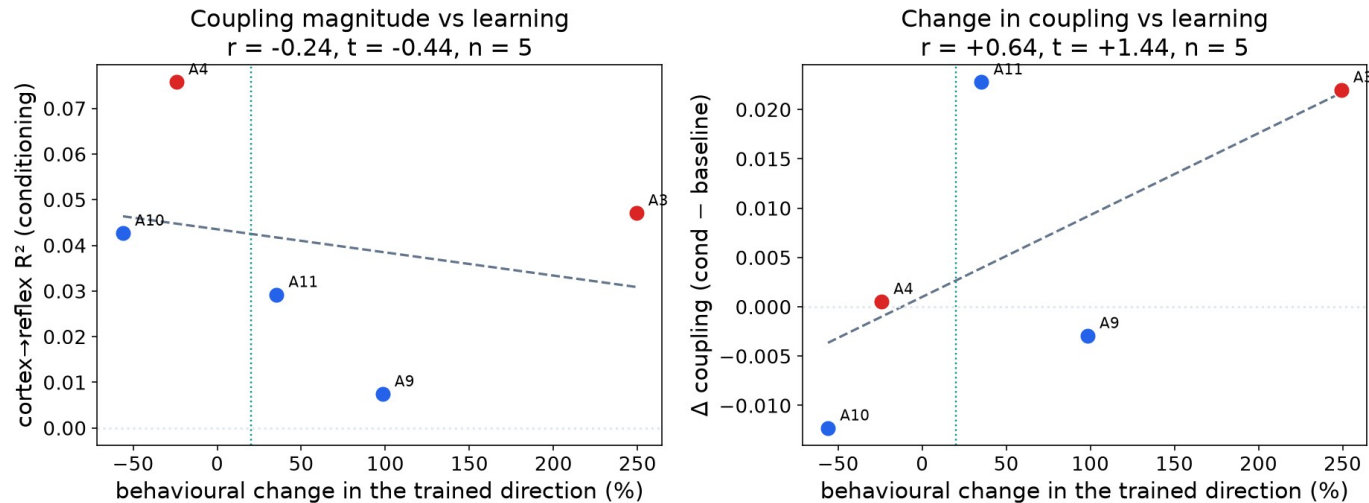
The test

Both channels record at once, so the cortical electrode could in principle be registering muscle activity. Muscle artefact lives at high frequency, so that is where contamination would appear.

Above 100 Hz: $r = +0.06$

Essentially zero, and the low bands are slightly negative. The cortical signal also marginally leads the muscle, which is the acceptable direction.

Does it scale with how much they learned?



$$r = +0.64$$

between how much an animal learned and how much its cortical coupling changed

The two animals that failed to learn show no increase — a built-in negative control.

With five animals this is a direction, not a result. Four more are next.

The metadata may be a measurement

An idea that came out of the last meeting, and one we think is worth pursuing.

What the reward threshold is

The experimenter raises it when the rat exceeds it and lowers it when the rat falls short of it. It is adjusted by hand, day by day, across the whole experiment.

It is, in effect, a staircase procedure.

Why that is interesting

It is a record of how hard the task was for that animal — its engagement and effort — written down independently of anything we measure from the electrodes.

No shared noise with our signal processing.

Right now our scaling result correlates two things measured from the same electrodes.

If cortical coupling instead tracks a behavioural record kept by hand at the time, that is a far harder result to dismiss. We would like to build this next.

Three things worth writing up

The method

A reproducible pipeline that recovers a twenty-year-old archive and learns a representation from it, on a laptop.

Useful to anyone with old chronic recordings sitting on a shelf.

The finding

Cortical activity carries a small, highly reliable single-trial signal about reflex size, independent of stimulus and excitability.

Extends Boulay et al. from a population relationship to a single-trial one.

The warning

The intuitive drift analysis produces a convincing result that three controls reject.

We would want to save the next group from spending months on it.

Target: a peer-reviewed machine-learning venue, framed as an extension of the existing cortical–spinal work.

Where a single-trial cortical readout could help



Closing the loop

If cortex predicts the reflex before the fact, conditioning protocols could adapt trial by trial rather than day by day.



Understanding who responds

Roughly a third of animals — and people — don't respond to conditioning. A cortical readout might show why, early.



Rehabilitation

Reflex conditioning is already used clinically after spinal cord injury. Any handle on the cortical side is a handle on individualising it.

All three depend on the effect being larger and more reliable than what we can currently demonstrate. We are not there yet — but this is the direction that would make it worth getting there.

What happens next

1

Process the remaining animals

Animals 7, 8, 13 and 14, all down-conditioned. Takes the sample from five to nine.

next

2

Power the scaling test

With nine animals the learning-versus-coupling correlation becomes a real test.

next

3

Frequency-resolved coupling

Which bands carry the reflex information? Sharpens the finding and the contamination argument.

next

4

The reward-criterion measure

An independent behavioural record to correlate against.

new thread

5

Write it up

Method, finding and warning.

after that

What we can say

SOLID

Conditioning reproduces

Both directions, after controlling for stimulus strength and muscle excitability. Three of five animals met the lab's criterion.

REAL

Cortex carries a single-trial signal

It predicts reflex size above its own shuffled control in 99 of 101 windows and in every animal. Small — about 2% of variance uniquely — but almost never absent.

USEFUL NEGATIVE

Average-state drift is not learning

The intuitive analysis produces a convincing result that three separate controls reject.

Every number above is corrected for stimulus intensity and pre-stimulus muscle activity, and survives a shuffle control.

What we know we haven't settled

1

Does the coupling grow with learning?

The correlation points the right way ($r = +0.64$) but five animals cannot settle it. Four more are next.

2

Which frequencies carry it?

Would sharpen both the finding and the argument against muscle contamination.

3

Is 2% of variance biologically meaningful?

It is reliable. Whether it matters is a question for this room more than for us.

4

What else could produce it?

We have excluded stimulus, muscle tone, drift and crosstalk. We would genuinely like to know what we are missing.

Code, data pipeline, figures and write-up



Project page

switchrao777.github.io/hroc-ncan

Findings, figures and links in one place.



Repository

github.com/switchrao777/hroc-ncan

The full pipeline. Two commands per animal, reproducible end to end.



Write-up

[docs/HROC_Report.pdf](#)

Six pages: background, methods, results, limitations, references.

What is in the repository

Every decoding and analysis script, the trained encoder, all 90+ figures, per-animal numerical results, a starter notebook, and setup instructions. Raw recordings are available through NCAN.

Thank you



With thanks to Dr. Jonathan Carp and Theresa Vaughan for their guidance throughout this project.

Suchith Rao • Tarun Senthil

switchrao777.github.io/hroc-ncan

Happy to take questions.